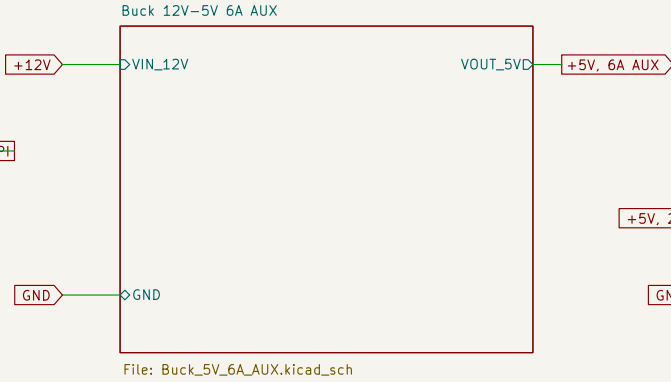
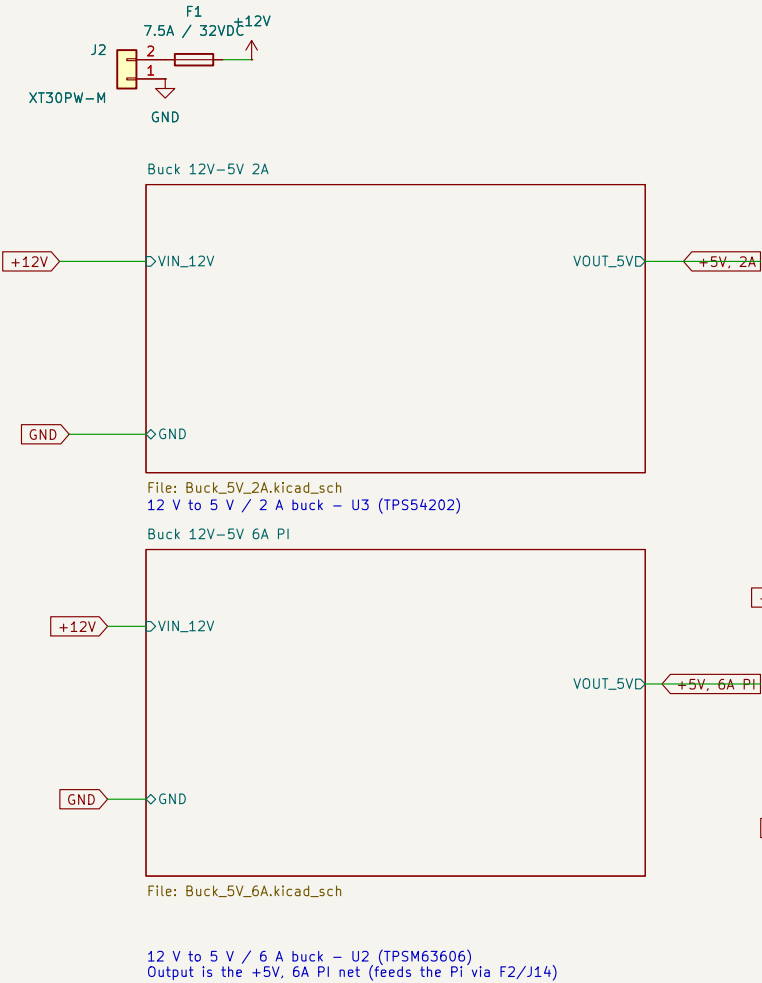
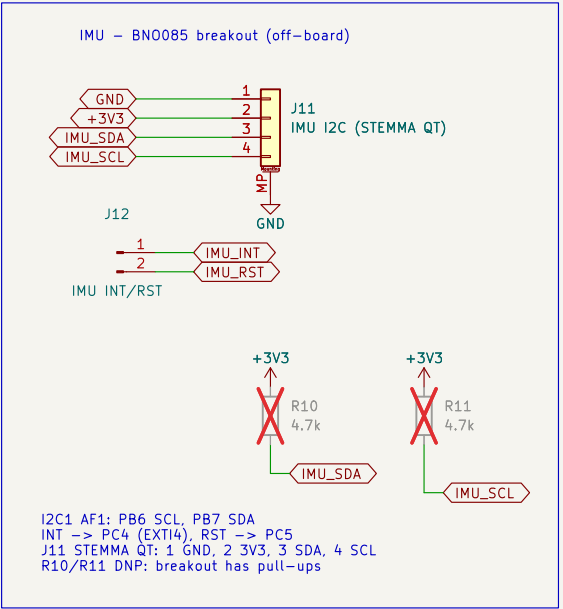
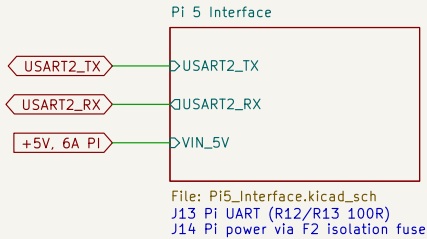
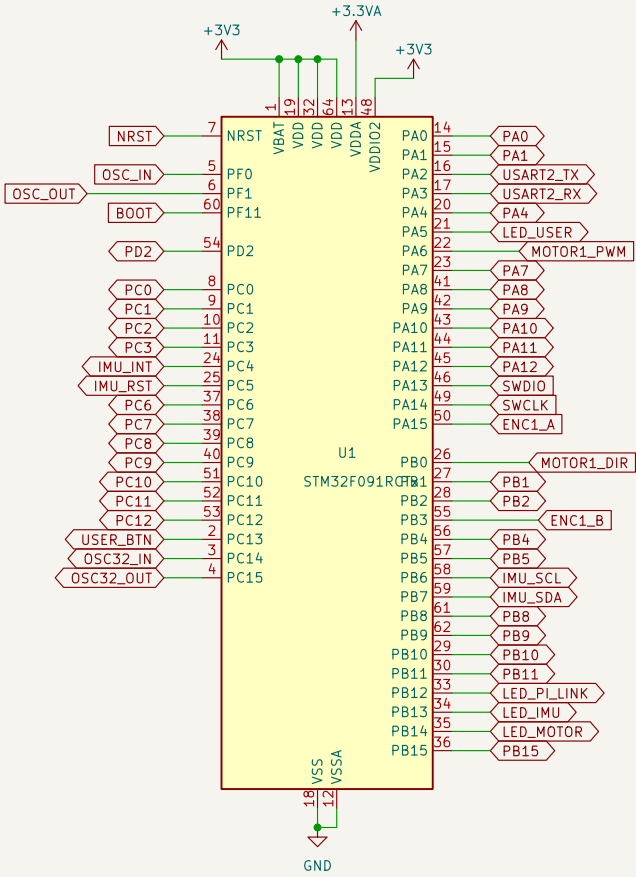
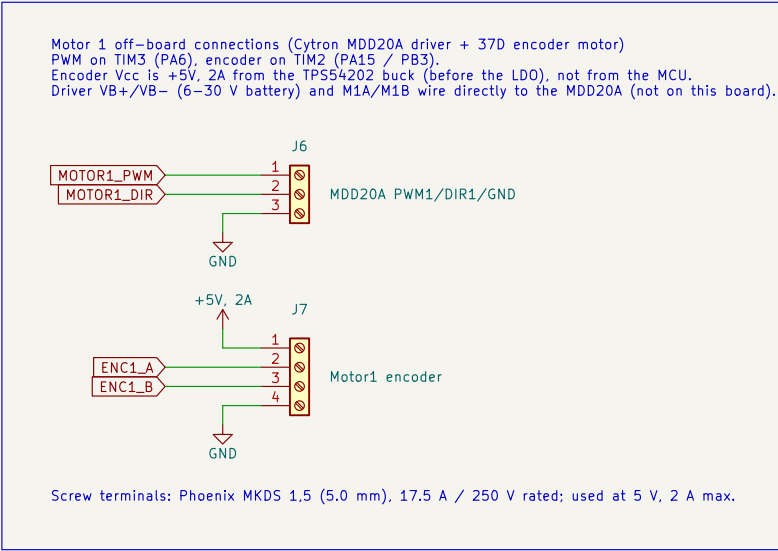
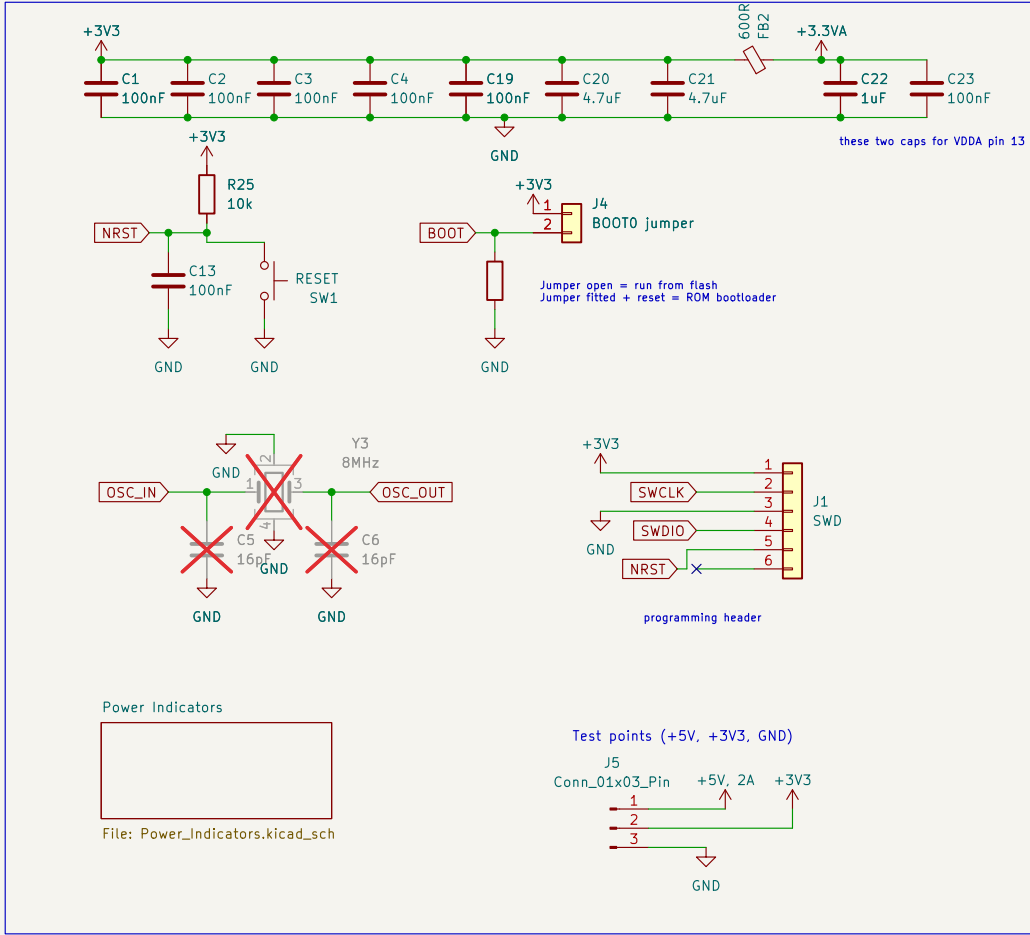
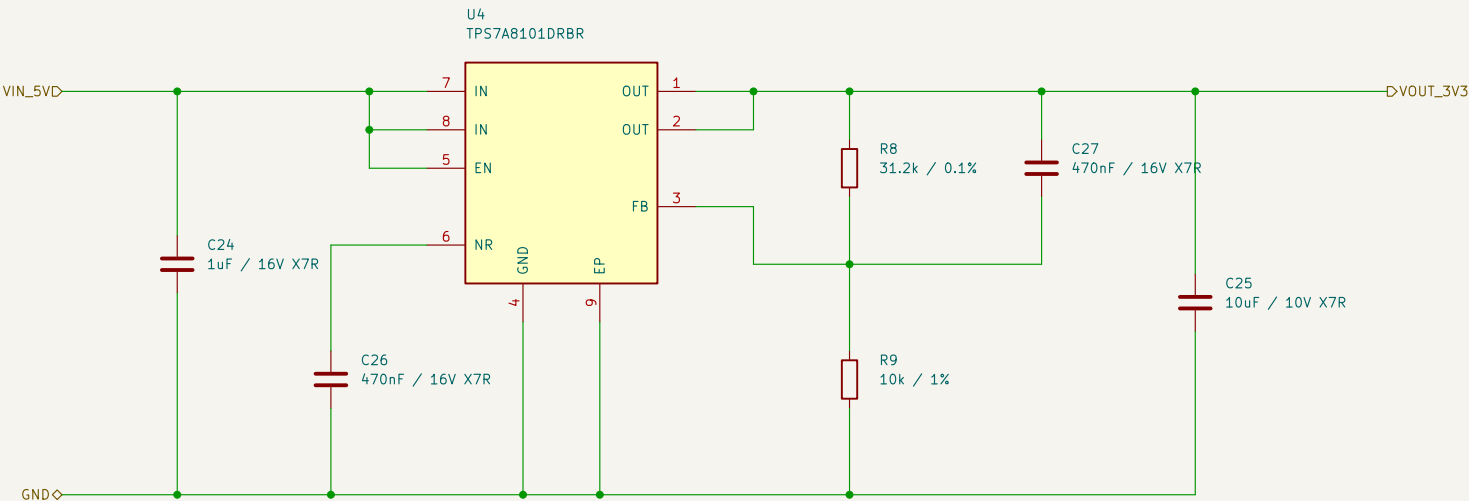


|                                                                                                  |                  |          |
|--------------------------------------------------------------------------------------------------|------------------|----------|
| Delivery robot senior design – MCU core<br>Derived from ST NUCLEO-64 MB1136 rev C-04, sheets 2-3 |                  |          |
| Sheet: /<br>File: delivery_robot_1.kicad_sch                                                     |                  |          |
| <b>Title: STM32F091RC core – ready for peripherals</b>                                           |                  |          |
| Size: A3                                                                                         | Date: 2026-09-24 | Rev: 1.0 |
| KiCad E.D.A. 10.0.6                                                                              |                  | Id: 1/8  |

# 5 V to 3.3 V LDO supply

Reusable sheet: VIN\_5V input, VOUT\_3V3 output, common GND



VOUT nominal =  $0.8 \times (1 + 31.2k / 10k) = 3.296 \text{ V}$   
1 A is the TOTAL output budget for STM32 + IMU + other 3.3 V loads.

C2: require  $\geq 4.7\mu\text{F}$  EFFECTIVE capacitance; selected part modeled at 8.8uF.  
EN tied to IN. C3 sets noise reduction / soft start; C4 bypasses R1.

Layout: capacitors close to U1; EP pad 9 to GND copper and thermal vias.  
Heat at 1 A: about 1.7 W at 5 V input; verify junction temperature on final PCB.

5 V motor / encoder supply branches BEFORE this sheet.  
Adafruit BNO085 breakout: 3.3 V rail to VIN; 3Vo is an output, leave separate.  
Bare BNO085: connect 3.3 V to VDD and VDDIO with its specified decoupling.

5 V to 3.3 V, 1 A total rail capacity

Senior Design

Sheet: /LDO Power/  
File: LDO\_3V3.kicad\_sch

Title: LDO power supply – reusable sheet

Size: A4

Date: 2026–10–05

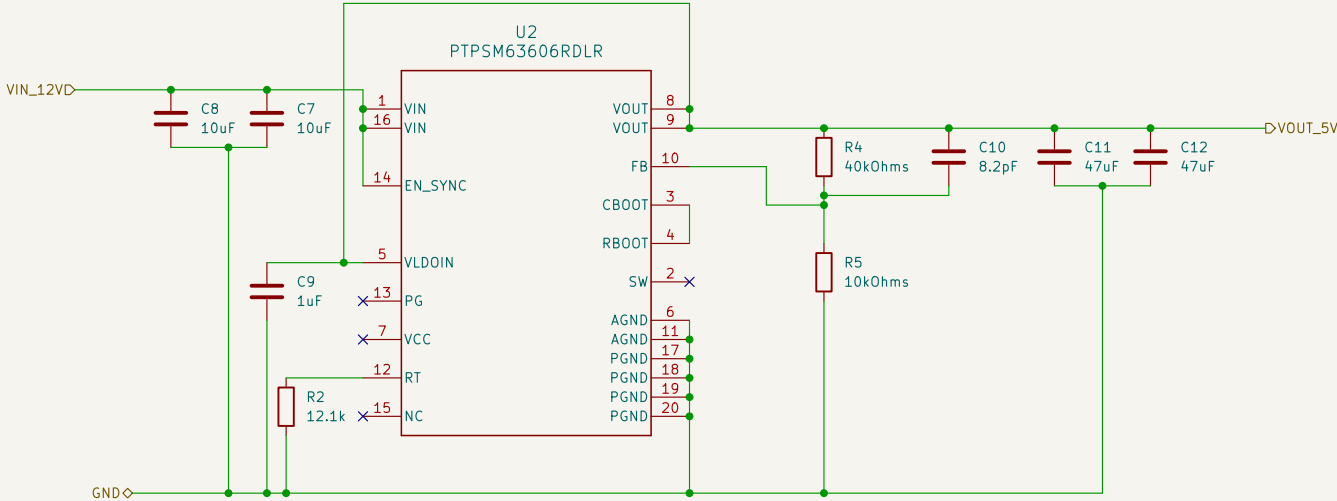
Rev: 1.0

KiCad E.D.A. 10.0.6

Id: 2/8

12 V – 5 V, 6A Buck Convertor

12 V to 5 V buck converter – TPSM63606 (U2)



Reusable sheet: VIN\_12V input, VOUT\_5V output, common GND  
VOUT set by R4/R5 divider on FB; RT = R2; EN tied to VIN.  
RBOOT shorted to CBOOT; SW left unconnected (inductor is inside the module).

12 V to 5 V buck, TPSM63606 (U2), 6 A rail

Senior Design

Sheet: /Buck 12V-5V 6A PI/

File: Buck\_5V\_6A.kicad\_sch

Title: Buck converter – reusable sheet

Size: A4

Date: 2026-10-06

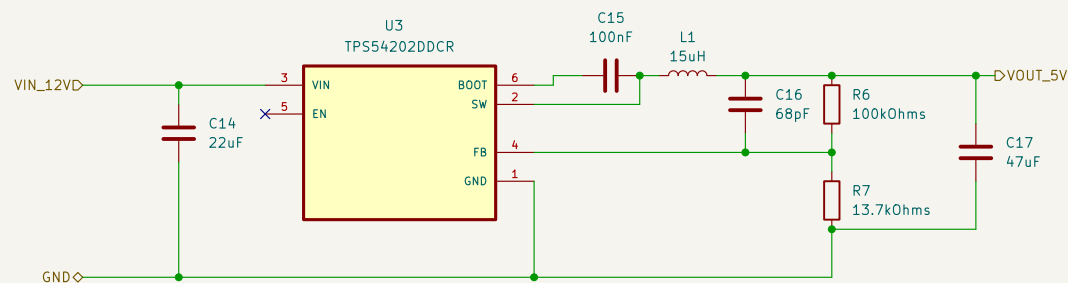
Rev: 1.0

KiCad E.D.A. 10.0.6

Id: 3/8

12 V – 5 V, 2A Buck Convertor

12 V to 5 V buck converter – TPS54202 (U3)



Reusable sheet: VIN\_12V input, VOUT\_5V output, common GND

12 V to 5 V buck, TPS54202 (U3), 2 A rail

Senior Design

Sheet: /Buck 12V-5V 2A/

File: Buck\_5V\_2A.kicad\_sch

**Title: Buck converter – reusable sheet**

Size: A4

Date: 2026-10-06

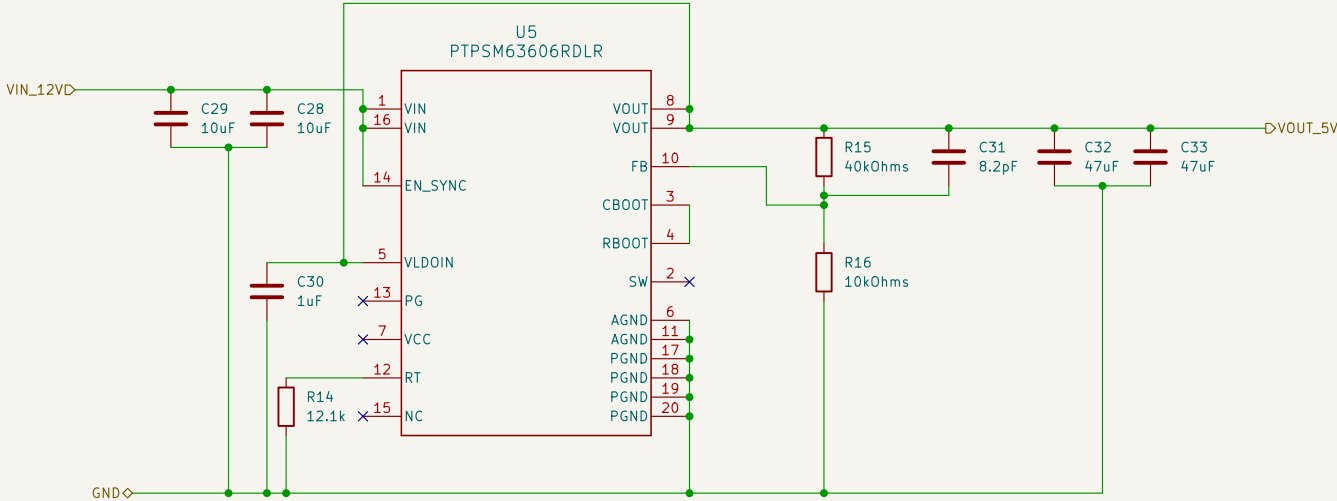
Rev: 1.0

KiCad E.D.A. 10.0.6

Id: 4/8

12 V – 5 V, 6A Buck Convertor

12 V to 5 V buck converter – TPSM63606 (U5)



Reusable sheet: VIN\_12V input, VOUT\_5V output, common GND.  
VOUT set by R15/R16 divider on FB; RT = R14; EN tied to VIN.  
RBOOT shorted to CBOOT; SW left unconnected (inductor is inside the module).

12 V to 5 V buck, TPSM63606 (U5), 6 A rail

Senior Design

Sheet: /Buck 12V-5V 6A AUX/

File: Buck\_5V\_6A\_AUX.kicad\_sch

Title: Buck converter – reusable sheet

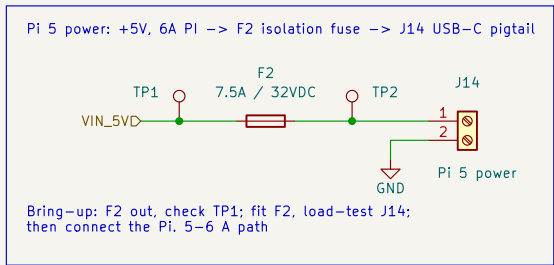
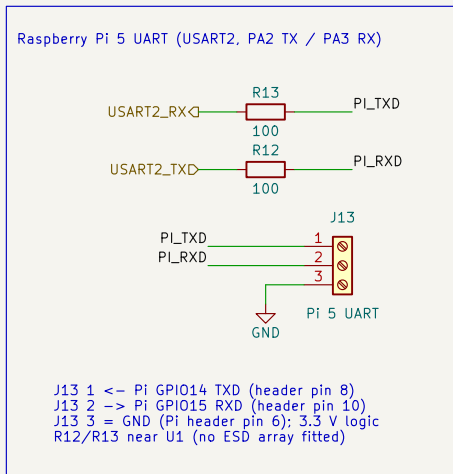
Size: A4

Date: 2026-10-06

Rev: 1.0

KiCad E.D.A. 10.0.6

Id: 5/8

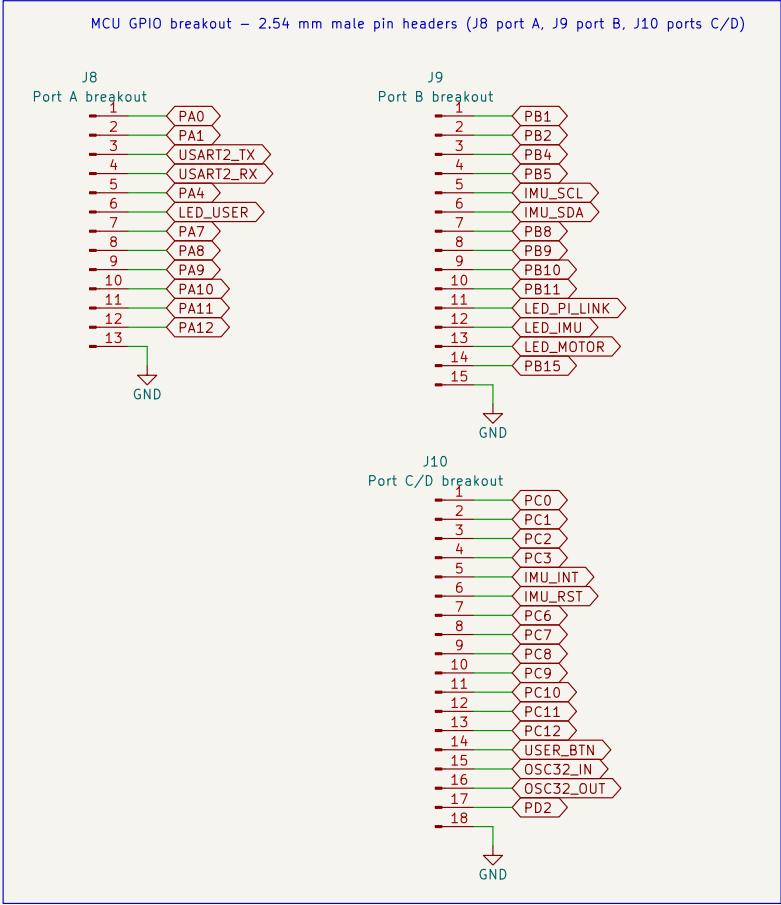


#### Senior Design

Sheet: /Pi 5 Interface/  
 File: Pi5\_Interface.kicad\_sch

#### Title: Raspberry Pi 5 interface – UART and power

|                     |                  |          |
|---------------------|------------------|----------|
| Size: A4            | Date: 2026-10-07 | Rev: 1.0 |
| KiCad E.D.A. 10.0.6 |                  | Id: 6/8  |



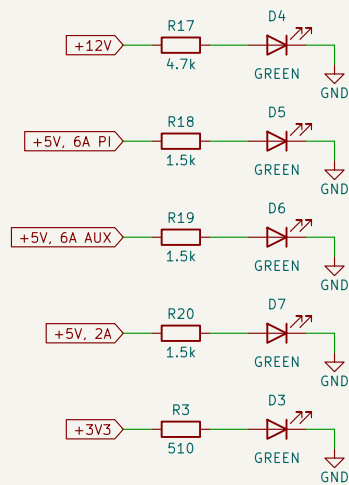
Senior Design

Sheet: /MCU GPIO Breakout/  
File: GPIO\_Breakout.kicad\_sch

Title: STM32 GPIO breakout headers

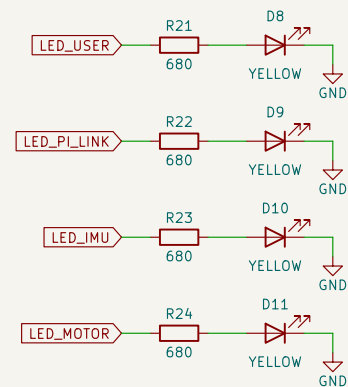
|                     |                  |          |
|---------------------|------------------|----------|
| Size: A4            | Date: 2026-10-07 | Rev: 1.0 |
| KiCad E.D.A. 10.0.6 |                  | Id: 7/8  |

Power rail indicators – one green LED per rail  
(Lite-On LTST-C190KGKT, 0603,  $V_f$  2.0 V typ)



LED current ~2–2.7 mA:  $R = (V_{rail} - 2.0 \text{ V}) / I$ . +12V uses 4.7k for the 10.8–14.6 V battery range.  
All resistors 0805; worst-case dissipation 34 mW (R17 at 14.6 V).

Debug LEDs – STM32 GPIO, active high (Lite-On LTST-C190KSKT yellow, 0603)



PA5 LED\_USER: heartbeat  
PB12 LED\_PI\_LINK: PI UART activity  
PB13 LED\_IMU: IMU data-ready  
PB14 LED\_MOTOR: motor running / fault  
680R:  $(3.3 - 2.1) / 680 = 1.8 \text{ mA per pin}$

#### Senior Design

Sheet: /Power Indicators/  
File: Power\_Indicators.kicad\_sch

#### Title: Power rail and debug indicator LEDs

|                     |                  |          |
|---------------------|------------------|----------|
| Size: A4            | Date: 2026-10-07 | Rev: 1.0 |
| KiCad E.D.A. 10.0.6 | Id: 8/8          |          |